

Poisson Profile Realizability and the Möbius Alignment Obstruction

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Abstract

The preceding TPC-209 stage identified a profile-aware nonprincipal-character problem after whole-frame Poisson reindexing. Its obstruction used arbitrary finite residue profiles, leaving open whether Schwartz regularity and the Poisson origin of those profiles might already force useful cross-divisor structure. We prove that they do not at finite modulus. For every prime $q > 2$, every vector in $\mathbb{C}^{\mathbb{F}_q^\times}$ is exactly the residue profile of a compactly supported smooth Fourier packet. The construction uses separated dual nodes and a single compactly supported smooth bump. We then insert the literal Möbius weights and realize the aligned family $B_D = \mu(D)U_D^*z$, obtaining an exact coherent-to-diagonal energy ratio equal to the number of squarefree unit divisors. Finally, we write the complete profile-aware energy as a positive-semidefinite cross-divisor Gram quadratic form. The result is a genuine admissible-profile obstruction: individual profile norms and Möbius signs cannot supply a saving. It is deliberately scoped; the independently chosen packets are not asserted to be the fully coupled physical twin-prime coefficients, so the actual physical profile bound, the strict $1/400$ gate, and the twin-prime endpoint remain open.

1 Introduction and claim ceiling

The current prime-dynamics route studies a fixed-gap prime-pair remainder in a standard-zero-hole, prime-modulus, reduced-residue variance. TPC-208 built the complete additive edge frame, and TPC-209 applied Poisson before any edge triangle. For one divisor D , the dual index

$$n = qr + kD$$

is shared by all additive vertices. Across divisors, however, the resulting residue packets are permuted by U_D . Multiplicative Fourier analysis gives a common character coordinate but leaves one profile per divisor [2].

The next natural question is narrower and more demanding:

Does the fact that each profile comes from a Schwartz function through the Poisson map, together with the literal Möbius coefficient $\mu(D)$, already prevent the alignment obstruction?

This paper answers the interface-level question negatively. The word “interface” is essential. The construction below allows a separate smooth Fourier packet for each divisor. It therefore proves that no theorem stated only in terms of the individual Schwartz/Poisson admissibility conditions

and the Mobius coefficient envelope can force cross-divisor cancellation. It does not prove that the literal, coupled TPC coefficient family realizes the same profiles.

The maximum justified claim is

PROVED_STRUCTURAL_L1 / STOP_SCOPED_PROFILE_CLASS.

More explicitly:

- finite residue-profile interpolation is proved exactly;
- a Mobius-weighted aligned family is proved exactly;
- the missing arithmetic input is identified as a cross-divisor Gram bound on the literal physical packets;
- no prime-only BDH estimate, Kloosterman attachment, arithmetic L^2 advance, fixed-atom credit, or twin-prime theorem is claimed.

2 The profile interface

Let $q > 2$ be prime and write

$$G_q = \mathbb{F}_q^\times, \quad N = q - 1, \quad P_q = I_N - \frac{1}{N} \mathbf{1}\mathbf{1}^*. \quad (2.1)$$

For a Schwartz function F_D on $\mathbb{R}$ and a unit divisor D modulo q , define the Poisson residue profile

$$B_{q,D}(s) = \sum_{\substack{n \in \mathbb{Z} \\ n \equiv s \pmod{q}}} \widehat{F}_D(n/q), \quad s \in G_q. \quad (2.2)$$

This is the dual packet appearing after the exact TPC-209 reindexing. The corresponding additive vector is $U_D B_{q,D}$, where

$$(U_D b)(k) = b(kD), \quad k \in G_q. \quad (2.3)$$

The TPC frame removes the constant direction. Thus the relevant profile energy is

$$\|P_q Y\|^2, \quad Y = \sum_{D \in \mathcal{D}} c_D U_D B_{q,D}. \quad (2.4)$$

The question is whether the range of (2.2) is sufficiently small to prevent coherent choices of the $B_{q,D}$.

3 Exact finite profile interpolation

Choose, for each $s \in G_q$, the dual integer

$$n_s = s + 10qs = s(10q + 1), \quad \rho_q = \frac{1}{4q}. \quad (3.1)$$

The points n_s/q are separated from one another by more than 10, while two distinct points in the same residue class are separated by exactly one. Let $\eta \in C_c^\infty((-1, 1))$ satisfy $\eta(0) = 1$.

Theorem 3.1 (finite profile interpolation). *For every target vector $b = (b(s))_{s \in G_q} \in \mathbb{C}^{G_q}$, define*

$$\hat{F}_b(\xi) = \sum_{s \in G_q} b(s) \eta\left(\frac{\xi - n_s/q}{\rho_q}\right). \quad (3.2)$$

Then $\hat{F}_b \in C_c^\infty(\mathbb{R})$, its inverse Fourier transform F_b is Schwartz, and

$$\sum_{\substack{n \in \mathbb{Z} \\ n \equiv t \pmod{q}}} \hat{F}_b(n/q) = b(t) \quad (t \in G_q). \quad (3.3)$$

Consequently, the profile map $F \mapsto B_q$ in (2.2) is onto $\mathbb{C}^{G_q}$, and any finite family of target profiles is realized by choosing the functions independently.

Proof. The support of the s -th summand in (3.2) is contained in $(n_s/q - \rho_q, n_s/q + \rho_q)$. Since $2\rho_q < 1$ and the points n_s/q are separated by more than 10, these intervals are disjoint and no interval contains any other point of the lattice $\{n/q : n \in \mathbb{Z}\}$. At the lattice point n_t/q , the t -th summand has value $b(t)$ and all other summands vanish. Every other point n/q in the same residue class is outside the support of every summand. Therefore the residue sum in (3.3) has exactly one nonzero term, equal to $b(t)$. A compactly supported smooth function has a Schwartz inverse Fourier transform, completing the proof. $\square$

Remark 3.2. *The theorem is finite and exact. It does not say that a single physical coefficient array can independently choose all divisor components. It says that smoothness and the Poisson residue map, by themselves, impose no such coupling.*

4 Mobius-weighted aligned profiles

Let $\mathcal{D}$ be a finite set of squarefree integers coprime to q , and put

$$c_D = \mu(D), \quad m = |\mathcal{D}|. \quad (4.1)$$

Take any centered vector $z \in \mathbf{1}^\perp$ and define

$$B_D = \mu(D) U_D^* z. \quad (4.2)$$

Theorem 3.1 realizes every B_D by a valid Schwartz packet.

Proposition 4.1 (exact Mobius alignment). *For the profiles (4.2),*

$$P_q \sum_{D \in \mathcal{D}} c_D U_D B_D = m z. \quad (4.3)$$

Moreover,

$$\sum_{D \in \mathcal{D}} |c_D|^2 \|P_q U_D B_D\|^2 = m \|z\|^2, \quad \left\| P_q \sum_{D \in \mathcal{D}} c_D U_D B_D \right\|^2 = m^2 \|z\|^2. \quad (4.4)$$

Thus the coherent-to-diagonal energy ratio is exactly m .

Proof. Since $U_D U_D^* = I$ and $\mu(D)^2 = 1$ on $\mathcal{D}$,

$$c_D U_D B_D = \mu(D) U_D (\mu(D) U_D^* z) = z.$$

Summing gives (4.3), and z is already in the range of P_q . Each individual output has squared norm $\|z\|^2$, which gives (4.4). $\square$

For the smallest nontrivial fixture, take $q = 5$, $\mathcal{D} = \{2, 3\}$, and $z = (1/2, -1/2, 0, 0)$. Both weights are -1 , yet the two weighted outputs are identical. The ratio in (4.4) is 2. This is the TPC-209 resonance with the additional fact that both profiles come from explicit compactly supported smooth Fourier packets.

5 The exact missing object: a cross-divisor Gram theorem

For arbitrary profiles define the output vectors

$$V_D = P_q U_D B_D, \quad G_{D,E} = \langle V_D, V_E \rangle. \quad (5.1)$$

Proposition 5.1 (Gram reduction). *The matrix $G = (G_{D,E})_{D,E \in \mathcal{D}}$ is Hermitian positive semidefinite and the whole-frame energy is exactly*

$$\left\| P_q \sum_D c_D U_D B_D \right\|^2 = \sum_{D,E} c_D \overline{c_E} G_{D,E}. \quad (5.2)$$

Its diagonal is

$$G_{D,D} = \|P_q B_D\|^2, \quad (5.3)$$

because U_D is unitary and commutes with P_q .

Proof. Equation (5.2) is the expansion of the squared norm. Any Gram matrix is Hermitian positive semidefinite. Finally, $U_D^* P_q U_D = P_q$, so (5.3) follows. $\square$

The diagonal data alone cannot imply cancellation. In the aligned family, $V_D = \mu(D)z$, so

$$G_{D,E} = \mu(D)\mu(E) \|z\|^2. \quad (5.4)$$

The matrix is rank one on the divisor index set, and the Mobius quadratic form in (5.2) adds every cross term constructively. Therefore the next legitimate arithmetic target is not a profile-norm estimate. It is a theorem that uses the literal physical coupling of the F_D to control the off-diagonal entries of G after all TPC normalizations are restored.

The same point can be expressed in multiplicative Fourier coordinates. If $A_D(\chi) = (\mathcal{M}B_D)(\chi)$, then

$$G_{D,E} = \sum_{\chi \neq \chi_0} \chi(D) \overline{\chi(E)} A_D(\chi) \overline{A_E(\chi)}. \quad (5.5)$$

Thus (5.5) is exactly the profile-aware nonprincipal-character interface identified in TPC-209, now with the admissible profile range made explicit.

Table 1: TPC-210 finite certificate.

quantity	value	status
prime moduli	3, 5, 7, 11, 13	exact
realized divisor-profile rows	20	exact
residue-coordinate rows	178	exact
support-geometry rows	34	exact
$q = 5$ Mobius divisors	2, 3	exact
$q = 5$ coherent-to-diagonal ratio	2	exact
continuous asymptotic estimate	none	not claimed

6 Finite certificate

The project certificate replays the interpolation geometry and the aligned family at $q = 3, 5, 7, 11, 13$. It uses exact rational arithmetic. The producer and independent checker agree on the residue nodes, support radius, Mobius weights, Gram matrices, and energy ratios.

The exact checks can be reproduced from the repository root with

```
cd papers/tpc-210-poisson-profile-realizability
PYTHONDONTWRITEBYTECODE=1 python -B experiments/run_certificate.py --check
PYTHONDONTWRITEBYTECODE=1 python -B experiments/independent_checker.py --check
PYTHONDONTWRITEBYTECODE=1 python -O -B experiments/independent_checker.py --check
PYTHONDONTWRITEBYTECODE=1 python -B experiments/profile_interpolation_sanity.py
```

The certificate is a finite QA artifact. It verifies the exact construction; it is not numerical evidence for an asymptotic prime correlation estimate.

7 Source boundary and route decision

TPC-209 left open a profile-aware prime-only estimate after the exact return to the V59 nonprincipal-character interface [1]. TPC-210 proves that the basic Schwartz/Poisson profile class is too large for a universal saving: it contains an exact Mobius-aligned family. Consequently, any future positive theorem must use additional physical information, such as a coupling between the divisor components, a common underlying coefficient array, or a source-backed cross-divisor Gram estimate.

The conclusion is a scoped stop:

```
PROFILE_CLASS_UNIVERSAL_SAVING=REFUTED_SCOPED,
ACTUAL_PHYSICAL_PROFILE_BOUND=OPEN,
FULL_GATE_B=OPEN, STRICT_1/400=UNPAID,
ARITHMETIC_ADVANCE=NO, L2=NONE.
```

8 Conclusion

The TPC-210 advance is a class-level separation theorem. Smooth Poisson packets can interpolate arbitrary finite residue profiles, and literal Mobius signs can be absorbed into adjoint dilations so that all whole-frame outputs align. The only route-compatible replacement is therefore a physical

cross-divisor theorem controlling the Gram matrix (5.1), with the exact zero-hole diagonal and all downstream prime-shell and block data retained.

This does not settle the twin-prime problem and does not negate the possibility that the actual coupled TPC packet has additional structure. It does remove a specific shortcut: profile regularity and Mobius signs alone are not that structure.

References

- [1] Liang Wang. Polarized local bdh scalar compiler, 2026. V59 session artifact, prime-dynamics theory repository.
- [2] Liang Wang. Whole-frame poisson reindexing and the mobius-dilation obstruction, 2026. TPC-209 session artifact, prime-dynamics theory repository.